THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** time period	Time period is measured.
2	**(d)** 0.005 mm	2 / 400 = 0.005 mm.
3	**(d)** rash seen by the doctor	Signs are observed.
4	**(b)** break in the circuit	Loose contact stops current.
5	**(b)** 4 N	Balanced forces.
6	**(b)** 600 Pa	240 / 0.4 = 600.
7	**(c)** particles move faster at higher temperature	Higher kinetic energy.
8	**(b)** O_2 and Fe	Oxygen and iron.
9	**(a)** 20 per cent	40/200 x 100.
10	**(b)** 65 degrees	i = 90 - 25 = 65.
11	**(b)** 45 degrees	Shadow = height.
12	**(d)** small fish	Third level.
13	**(c)** less than 1 per cent of the radius	35/6400 is about 0.5 per cent.
14	**(a)** 20 degrees C per minute	Rise of 20 each minute.
15	**(c)** has similar cells doing one job	Organs have more than one tissue type.
16	**(d)** entry, multiplication, symptoms	Pathogen enters and multiplies before symptoms.
17	**(b)** heating	All use resistive heating.
18	**(d)** positive charge	Like charges repel.
19	**(d)** more air is pumped in	More collisions.
20	**(d)** gives out extra heat when it condenses on the skin	Condensation of steam releases the heat absorbed during vaporisation.
21	**(b)** sulphur	Sulphur is a non-metal; mercury is a metal that is liquid at room temperature.
22	**(b)** colloid	Tyndall effect.
23	**(d)** 20 cm	R = 2f.
24	**(b)** 354	12 x 29.5.
25	**(a)** oxygen	Photosynthesis.
26	**(b)** 105 L	21/100 x 500.
27	**(b)** check the experimental method	Several outliers suggest a faulty procedure.
28	**(c)** provide contrast	Stains help visibility.
29	**(c)** white blood cells and antibodies	Defensive cells and proteins.
30	**(d)** compass needle deflection	Oersted.
31	**(a)** change in colour	Force does not change colour.
32	**(b)** the same type of storm with different names	Tropical cyclones.
33	**(a)** liquid	Liquefaction.
34	**(a)** sonorous and hard	Alloy property.
35	**(d)** more concentrated	More solute.
36	**(b)** principal focus	Focus.
37	**(a)** west after sunset	Young Moon.
38	**(b)** organic waste rots in it	Decomposition uses oxygen.
39	**(d)** 3 per cent	Most is salty.
40	**(b)** length of the string	Only mass is changed.
41	**(b)** a leaf	Only plant cells have all three.

42	**(c)** contaminated drinking water	Flood mixes sewage with water.
43	**(b)** nichrome	Higher resistance.
44	**(b)** 8 N	20 - 12 = 8 N.
45	**(c)** 500 Pa	20/0.04 = 500 Pa.
46	**(c)** particles of salt fit in the gaps between water particles	Spaces between particles.
47	**(c)** NH_3 and CH_4	Ammonia and methane.
48	**(a)** 30 g	0.12 x 250.
49	**(d)** 40 degrees	i + r = 80.
50	**(c)** to the south of the observer	Shadow is opposite to the Sun.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** foxes are removed	Fewer predators.
2	**(a)** a thick CO2 atmosphere	Runaway greenhouse.
3	**(d)** cm	Height is length.
4	**(c)** 10x	400 / 40 = 10.
5	**(c)** before exposure to disease	Preventive.
6	**(b)** the current is switched off	Electromagnet demagnetises.
7	**(a)** it induces opposite charge near it	Charge separation.
8	**(b)** outside pressure is lower than inside	Lower external pressure.
9	**(d)** boiling point (condensation point)	Condensation.
10	**(a)** copper	Copper is an element; the others are mixtures of metals.
11	**(a)** milk	Colloid.
12	**(b)** convex mirror	A convex mirror gives a wide field of view.
13	**(b)** 11 days	365 - 354.
14	**(a)** global warming	Greenhouse effect.
15	**(a)** extremely high pressure	Pressure keeps it solid.
16	**(d)** height and time	Height changed, time measured.
17	**(c)** 10 mm	0.1 x 100 = 10 mm.
18	**(c)** a physical barrier to germs	Barrier.
19	**(c)** coin, key and iron nail	Metals.
20	**(b)** 0.3 kg	3/10.
21	**(c)** over land under thunderclouds	From severe thunderstorms.
22	**(d)** evaporates quickly	Takes heat.
23	**(d)** a good conductor and ductile	Properties.
24	**(b)** temperature and amount of sugar	Constants.
25	**(a)** virtual, erect and smaller	Always.
26	**(d)** sunset	Opposite to the Sun.
27	**(b)** increase	Fewer predators.
28	**(c)** 156 L	78/100 x 200.
29	**(d)** 12.0 s	(12.1 + 11.9 + 12.0)/3 = 12.0.
30	**(b)** an onion bulb	Underground bulb cells lack chloroplasts but have cell walls.
31	**(c)** stagnant	They breed in still water.
32	**(a)** the switch is open	No current.
33	**(c)** 0 N	Balanced.
34	**(b)** 200 Pa	20/0.1 = 200 Pa.
35	**(c)** solid, liquid, gas	Solids have the least gaps.
36	**(d)** compound	New substance.
37	**(a)** 475 g	Salt = 25 g.
38	**(d)** 4 m/s	Relative speed.
39	**(b)** 15 days	Half a month.
40	**(a)** producer level	Energy decreases at each step.
41	**(d)** 30	3 per cent.
42	**(b)** thickness of dough	It is kept same.

43	**(b)** They are large and store water and nutrients	Vacuoles in plants are large.
44	**(c)** monitoring physical health	Physical measurements.
45	**(c)** glows brighter	Greater voltage.
46	**(b)** charge flows to earth	Grounded.
47	**(d)** land is warmer than sea in the daytime	Air rises over land.
48	**(a)** water at 90 degrees Celsius	Higher temperature.
49	**(d)** silicon	Metalloid features.
50	**(b)** sand in water	Suspension.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** virtual, erect and enlarged	Magnifying mirror.
2	**(a)** 33 days	3 x 11.
3	**(d)** burning of fossil fuels and deforestation	Human activity.
4	**(c)** is farther from the Sun and has a thin atmosphere	Weak greenhouse.
5	**(a)** type of soil	Only the soil type is changed.
6	**(a)** nucleus and cytoplasm	Common parts.
7	**(d)** trapping dust and germs	Sticky layer.
8	**(b)** circles around the wire	Concentric circles.
9	**(c)** have two poles	Poles always occur in pairs.
10	**(c)** eye wall	Around the eye.
11	**(d)** vapour cannot escape; condensation balances evaporation	Equilibrium.
12	**(d)** good conductors of heat	Heat conduction.
13	**(c)** 4 g remains undissolved	Saturation reached.
14	**(d)** at the focus	Parallel rays.
15	**(a)** sunrise	Near the Sun.
16	**(b)** stray dogs and diseases	Carcasses remain.
17	**(d)** seismic waves of earthquakes	Waves travel through layers.
18	**(c)** 1.8 s	36/20 = 1.8 s.
19	**(c)** They have genetic material but no true nucleus.	Prokaryotic cells.
20	**(a)** immunity	The body remembers the pathogen.
21	**(c)** more clips	More turns, stronger.
22	**(a)** 5 N upward	Balanced forces.
23	**(d)** 30000 Pa	600/0.02 = 30000 Pa.
24	**(a)** diffusion in air takes time	Particles travel gradually.
25	**(c)** 1 : 8	2 g H : 16 g O.
26	**(b)** 10 per cent	15/150 x 100.
27	**(b)** 2 m	2 x 1.0.
28	**(a)** 15 January	Half a lunar month.
29	**(d)** more than one trophic level	Omnivore.
30	**(b)** air is thinner with lower oxygen amount per breath	Pressure falls.
31	**(a)** unreliable	Repeatability gives reliability.
32	**(b)** protozoan	Protozoa have a nucleus.
33	**(c)** type 2 diabetes	Linked to diet and exercise.
34	**(a)** heats less due to low resistance	Low resistance.
35	**(a)** open	Induction.
36	**(b)** rising	Warm air rises.
37	**(c)** evaporation	Liquid to vapour.
38	**(b)** graphite (carbon)	Exception.

39	**(b)** a suspension	Large particles.
40	**(b)** principal focus	Focus.
41	**(a)** Adhik maas	Corrects the gap.
42	**(b)** stomata	Transpiration.
43	**(b)** fossil groundwater in deep layers	Replenished very slowly.
44	**(a)** How long does ice take to melt in a warm room?	Time must be measured.
45	**(d)** contains genetic material	It is the genetic region.
46	**(a)** resistant bacteria	Bacteria evolve.
47	**(b)** melt as 17 A exceeds 15 A	2+5+6+4 = 17 A.
48	**(b)** a non-contact force	Acts without touching.
49	**(d)** air sinks at the centre	Sinking air.
50	**(a)** increased pressure raises the boiling point	High pressure.

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